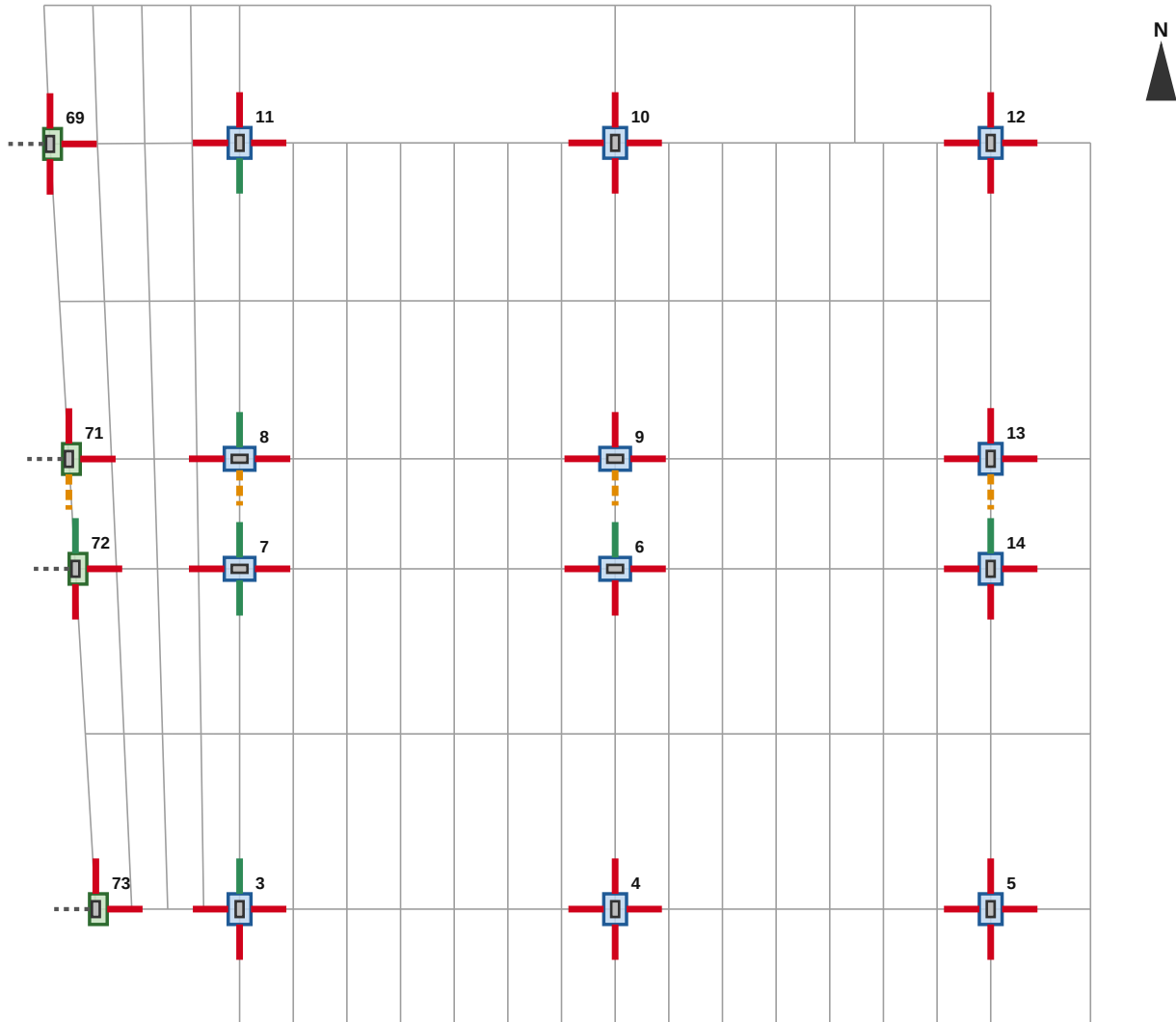


Collar details - double collar moment joints

Plates 12 mm (cap) and 12 mm (shelf), A36. Strip beyond the column: 95 mm where a beam laps ($L_{lap} 80 + gap 15$), 30 mm at the building edge. Joint classes from joint_classes.m; beams on each side from the ETABS model. Sides are global: W = -X, N = +Y, E = +X, S = -Y.

Key plan



Plates coloured by type. Beam ends: red M (moment, lapped on cap and shelf), green S (shear, on the shelf), orange dashed NL (below the collar, shear tab), grey dotted E (edge, 30 mm strip). (printed at about 1:100)

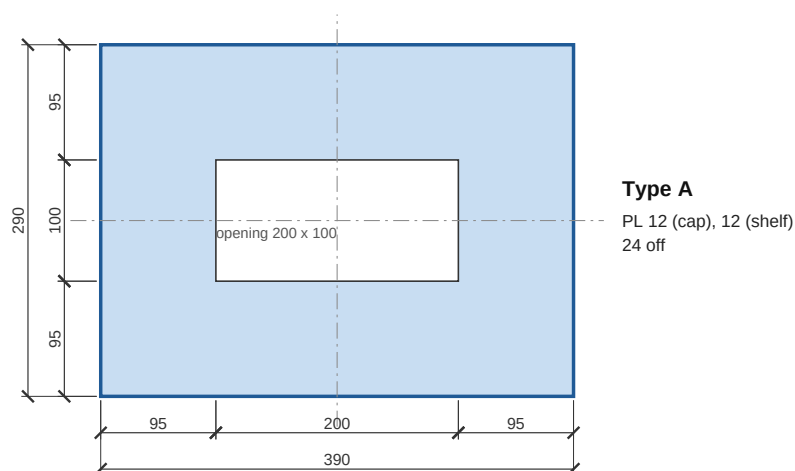
Joint schedule

Joint	Column	Strong axis	W N E S	Plate	Placement	S beams (shelf)	NL beams (tab)
3	43	Y	M S M M	A	long side along Y	18	-
4	44	Y	M M M M	A	long side along Y	-	-
5	45	Y	M M M M	A	long side along Y	-	-
6	46	X	M S M M	A	long side along X	120	-
7	47	X	M S M S	A	long side along X	100, 18	-
8	48	X	M S M NL	A	long side along X	26	100
9	49	X	M M M NL	A	long side along X	-	120
10	50	Y	M M M M	A	long side along Y	-	-
11	51	Y	M M M S	A	long side along Y	26	-
12	52	Y	M M M M	A	long side along Y	-	-
13	53	Y	M M M NL	A	long side along Y	-	70
14	54	Y	M S M M	A	long side along Y	70	-
69	103	Y	E M M M	B	long side along Y, 30 mm strip to W	-	-
71	104	Y	E M M NL	B	long side along Y, 30 mm strip to W	-	25
72	105	Y	E S M M	B	long side along Y, 30 mm strip to W	25	-
73	106	Y	E M M N	B	long side along Y, 30 mm strip to W	-	-

Strong axis: global axis of the column's 200 mm side, where its strong beams run. Placement: how the plate outline sits on plan.

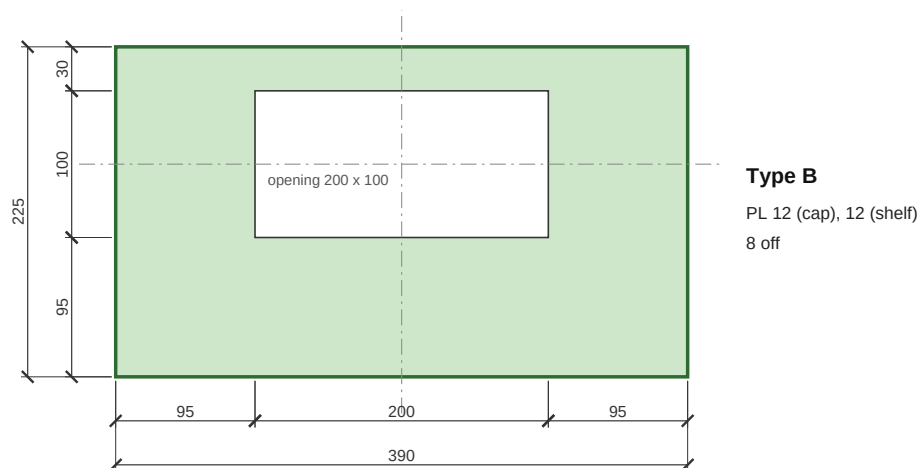
Plate types

Type A: PL 390 x 290, opening 200 x 100, cap 12 mm and shelf 12 mm, 12 joints, 24 plates



Joints 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14. The opening is the column outline; the fabricator adds the fit-up clearance. (printed at about 1:7.5)

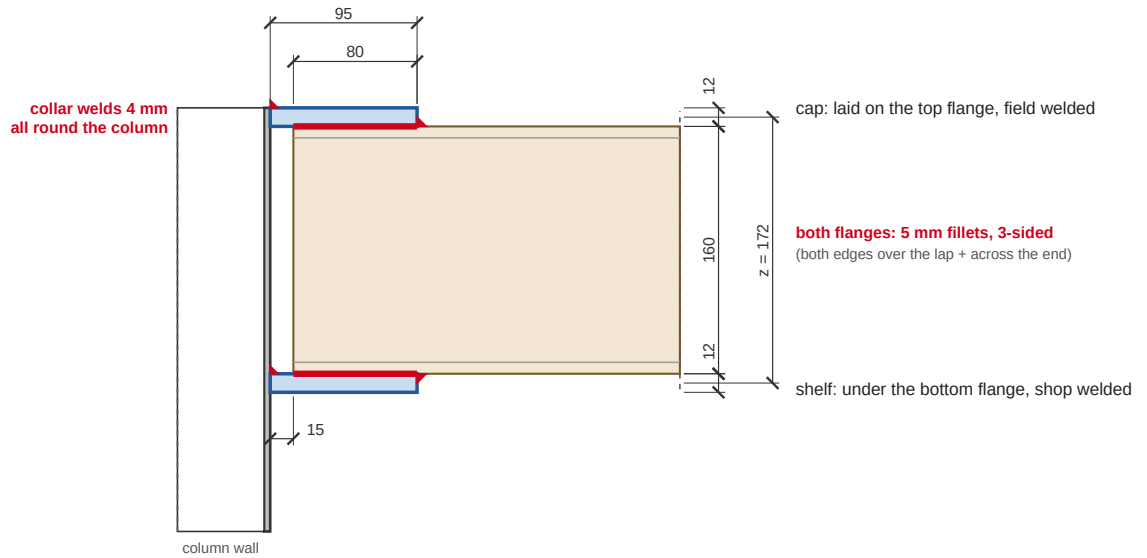
Type B: PL 390 x 225, opening 200 x 100, cap 12 mm and shelf 12 mm, 4 joints, 8 plates



Joints 69, 71, 72, 73. The opening is the column outline; the fabricator adds the fit-up clearance. (printed at about 1:5)

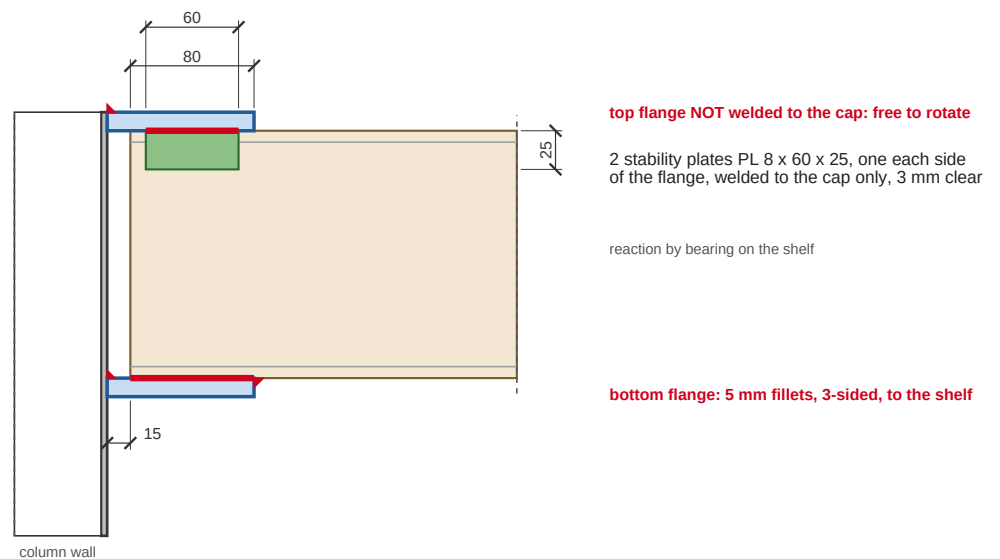
Connection types, section along the beam

M - moment beam: both flanges lapped and welded to the cap and the shelf



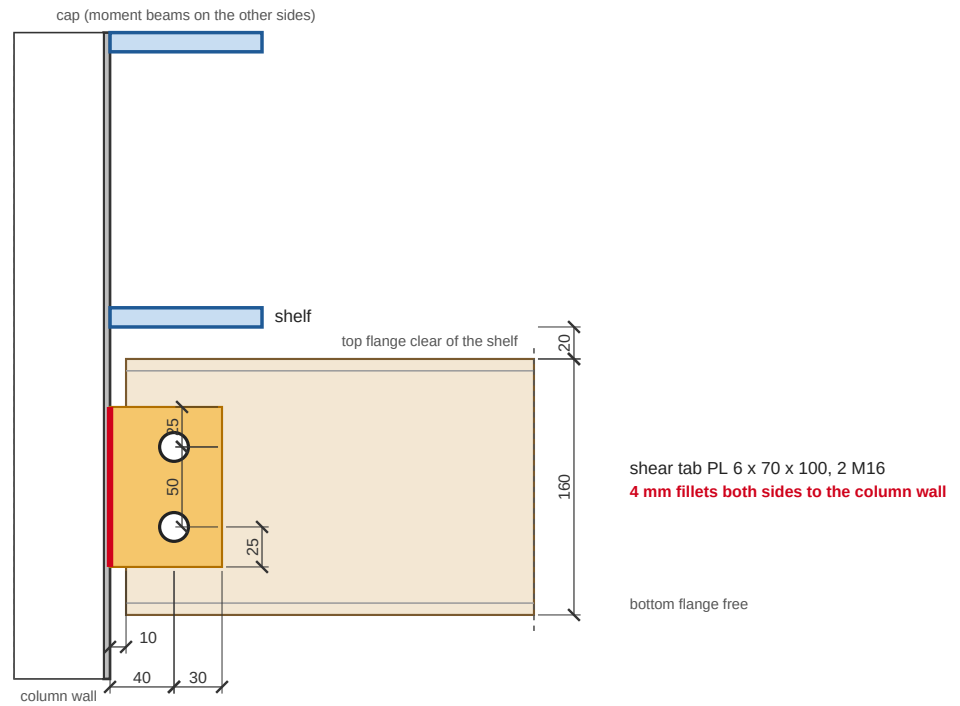
(printed at about 1:5)

S - shear only: on the shelf, top flange free under the cap



(printed at about 1:5)

NL - beam below the collar: shear tab welded to the column wall, bolted to the web

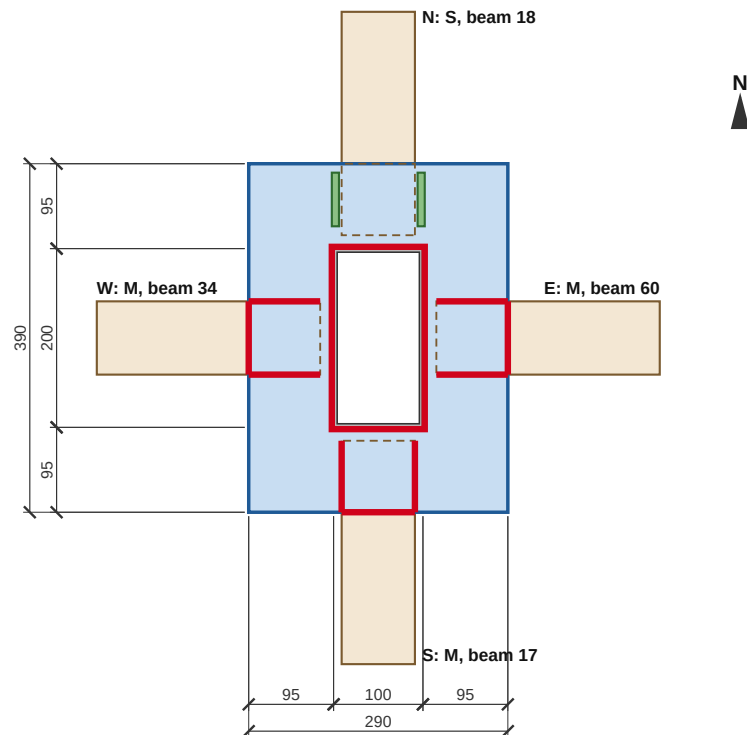


(printed at about 1:5)

Shear tab: PL 6, 2 M16 bolts in 18 mm holes, pitch 50, edge 25 (vertical) and 30 (horizontal), bolt line 40 mm from the wall, beam end 10 mm from the wall, 4 mm fillets both sides. Checked as P1 to P11 in dmj_lib.

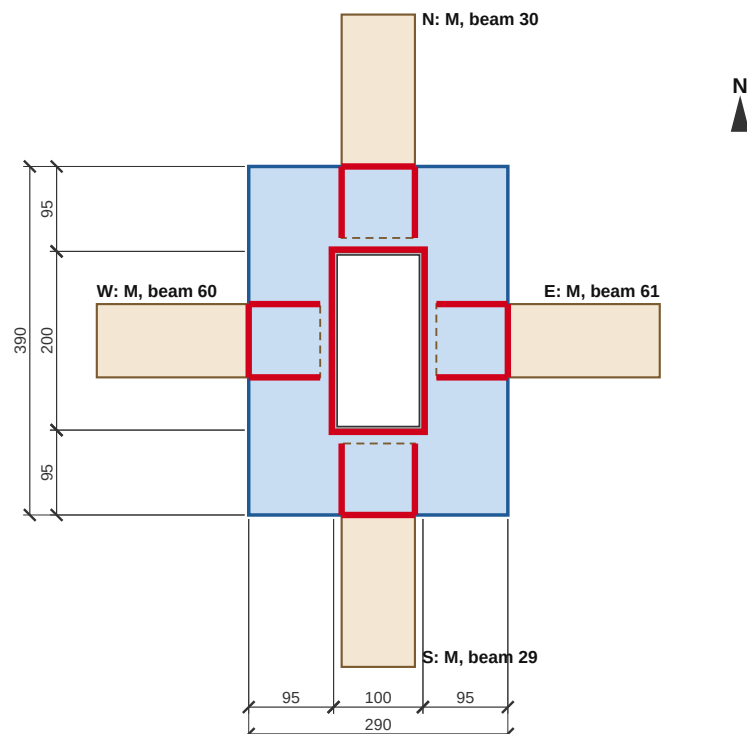
Joint plans (cap level, seen from above)

Joint 3 - type A - class M S M M - column 43, strong axis Y



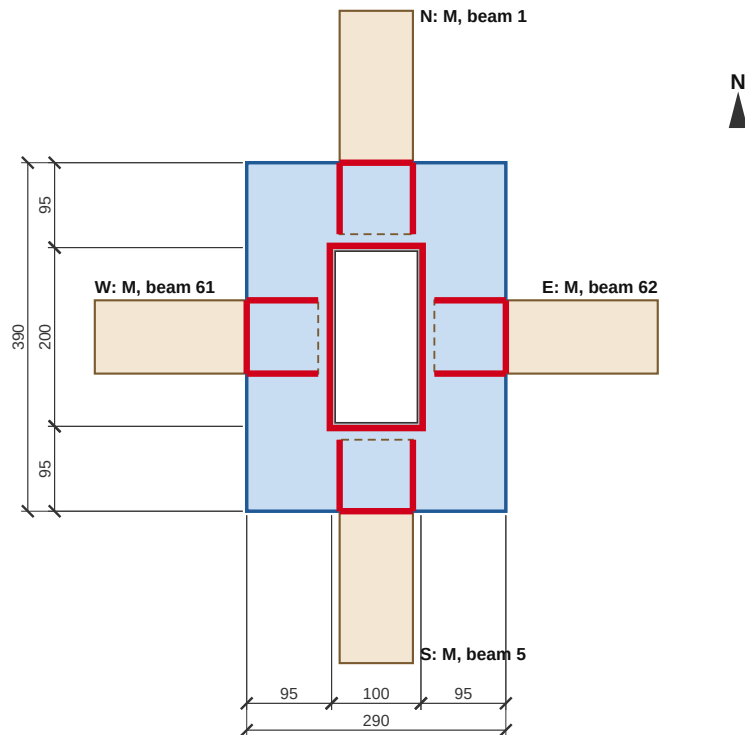
Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab. (printed at about 1:7.5)

Joint 4 - type A - class M M M M - column 44, strong axis Y



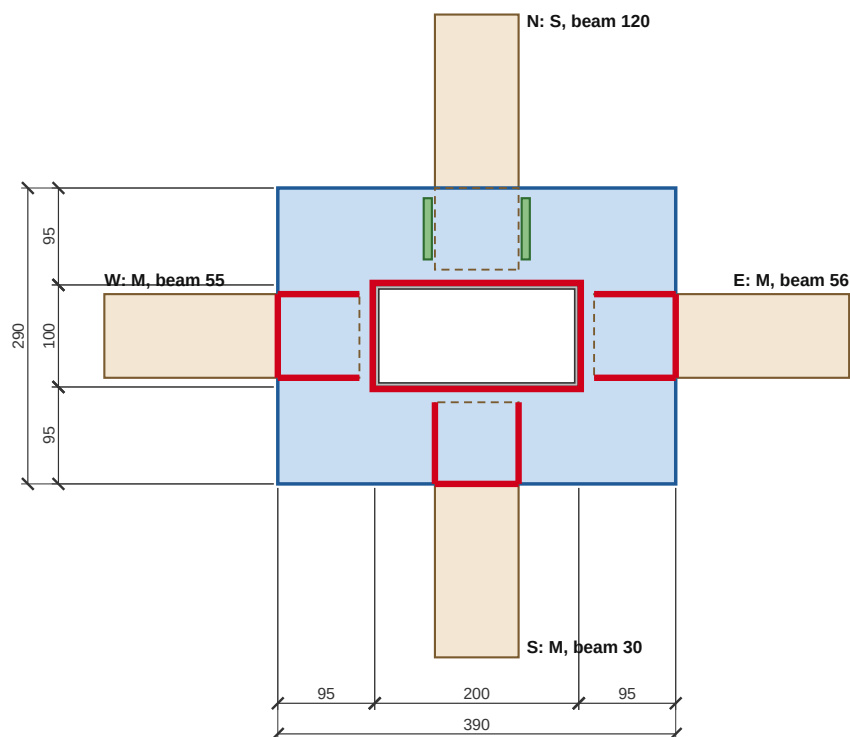
Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab. (printed at about 1:7.5)

Joint 5 - type A - class M M M M - column 45, strong axis Y



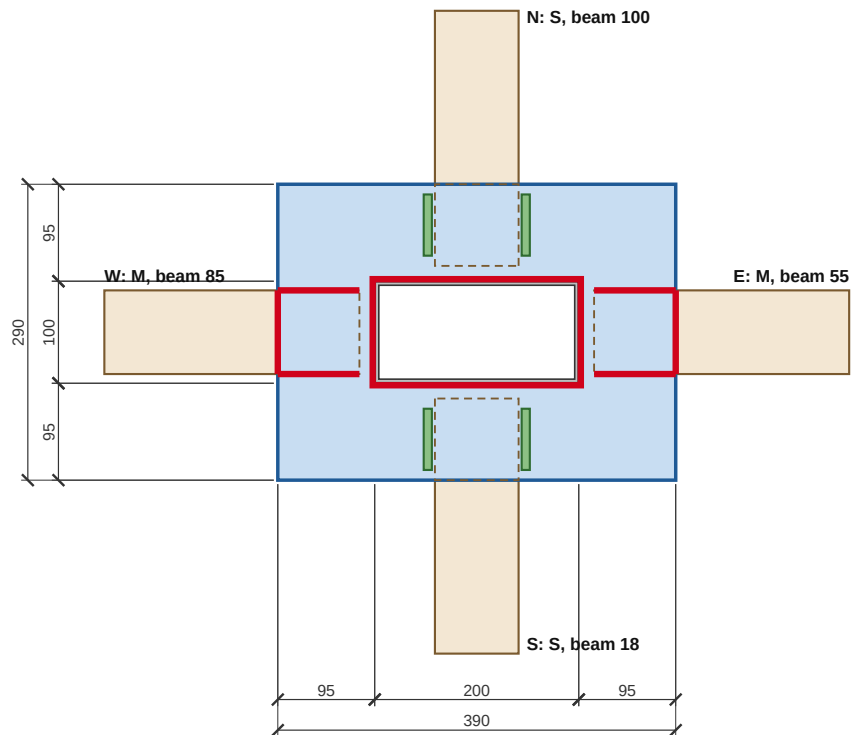
Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab. (printed at about 1:7.5)

Joint 6 - type A - class M S M M - column 46, strong axis X



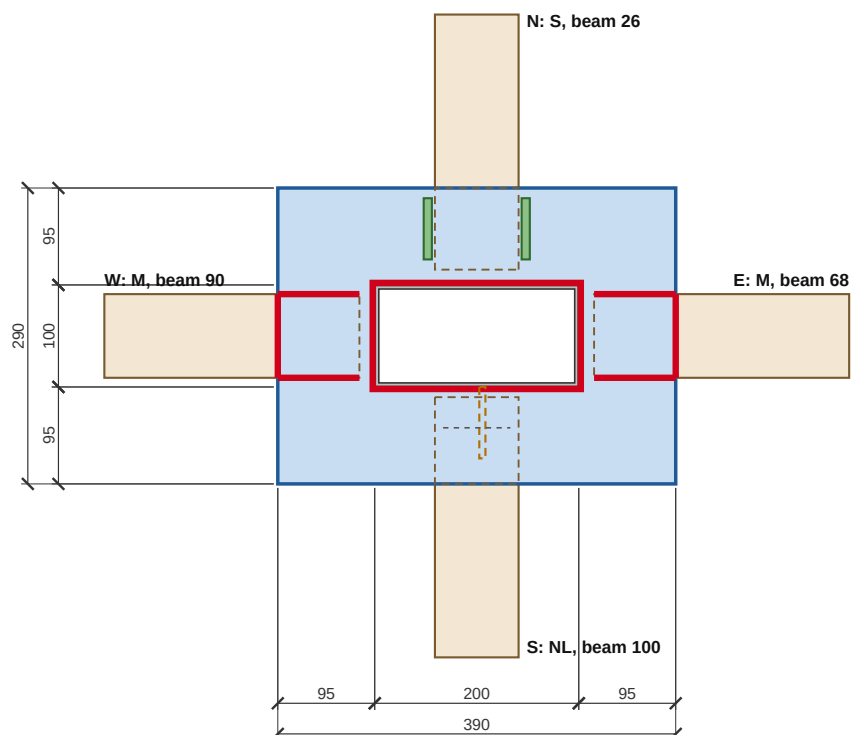
Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab. (printed at about 1:7.5)

Joint 7 - type A - class M S M S - column 47, strong axis X



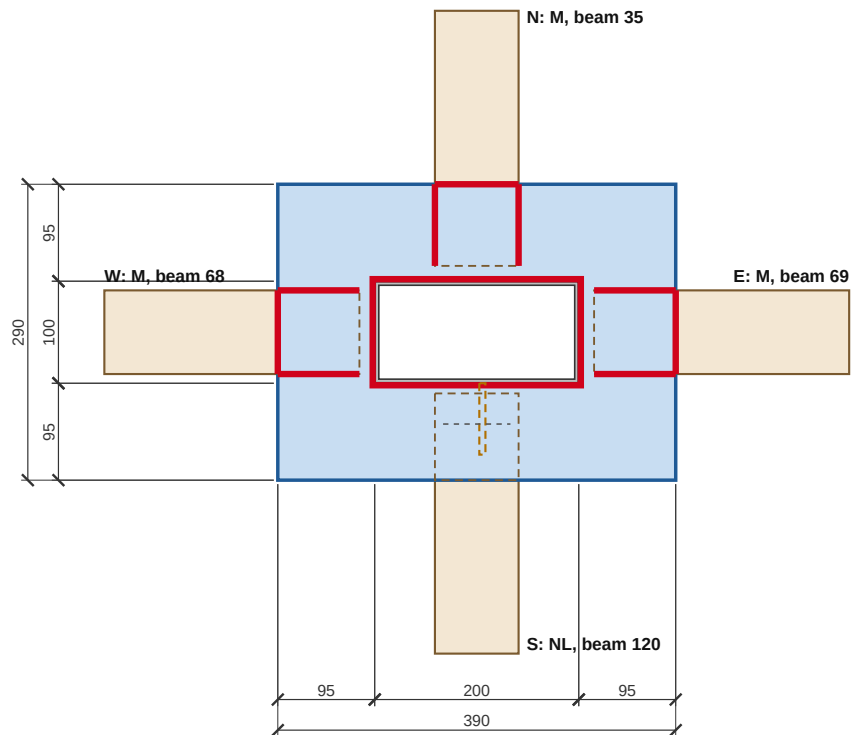
Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab.
(printed at about 1:7.5)

Joint 8 - type A - class M S M NL - column 48, strong axis X



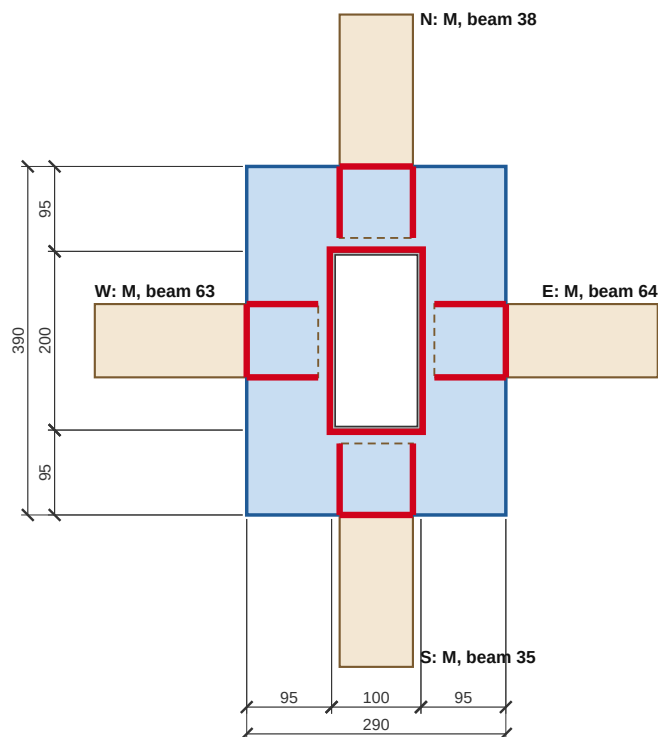
Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab.
(printed at about 1:7.5)

Joint 9 - type A - class M M M NL - column 49, strong axis X



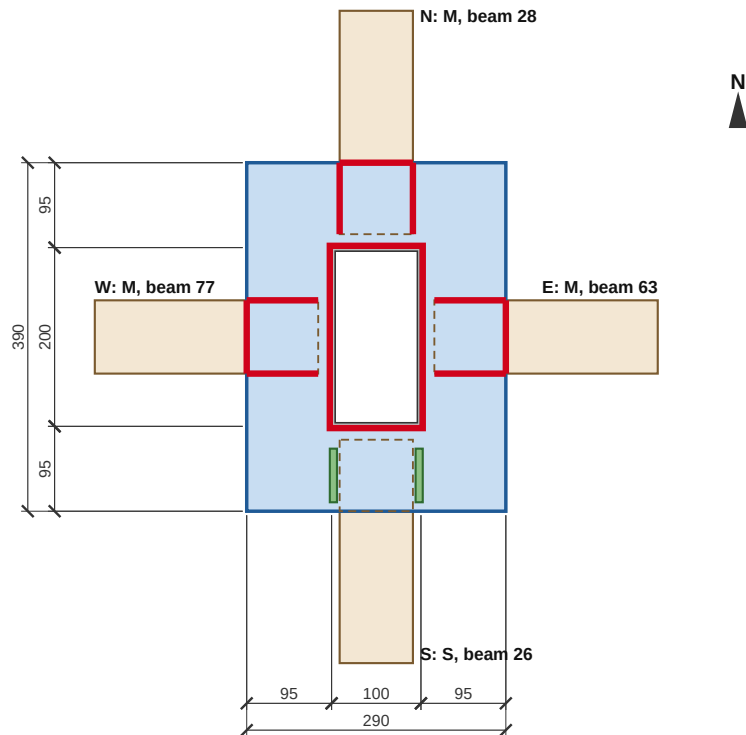
Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab.
(printed at about 1:7.5)

Joint 10 - type A - class M M M M - column 50, strong axis Y



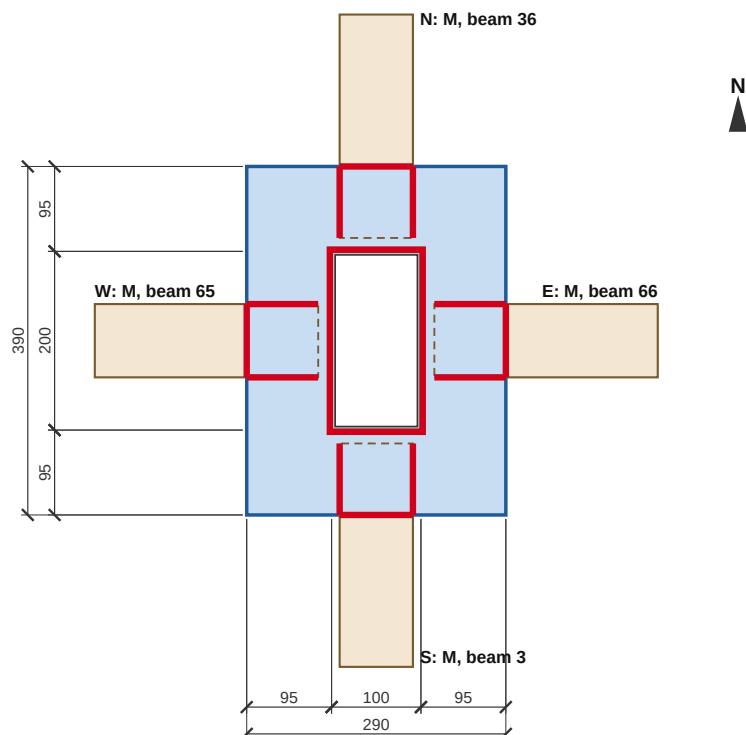
Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab.
(printed at about 1:7.5)

Joint 11 - type A - class M M M S - column 51, strong axis Y



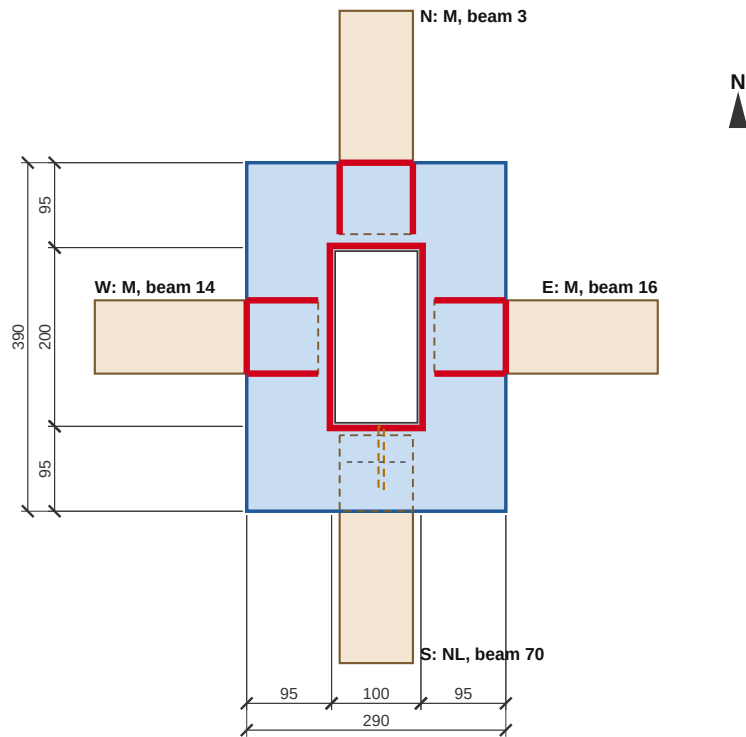
Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab.
(printed at about 1:7.5)

Joint 12 - type A - class M M M M - column 52, strong axis Y



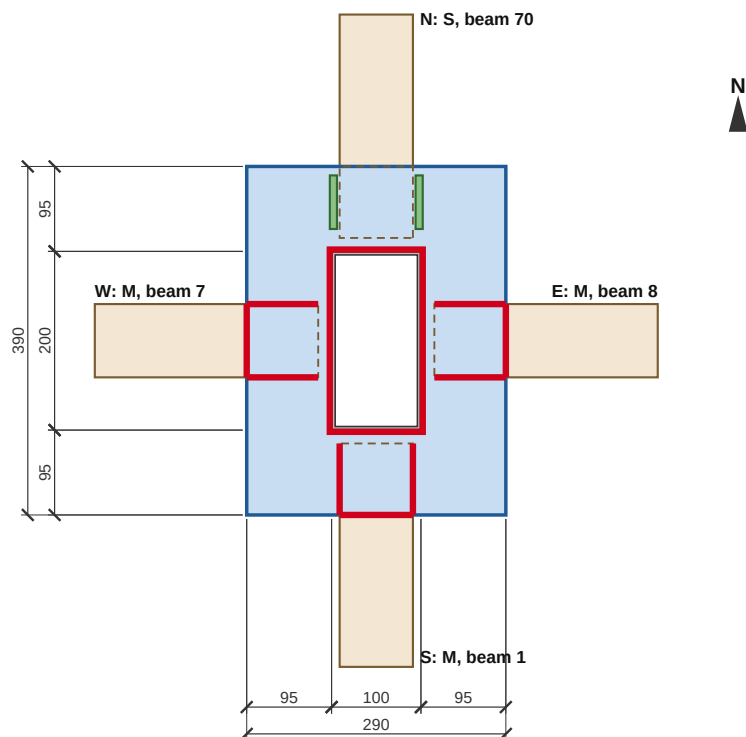
Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab.
(printed at about 1:7.5)

Joint 13 - type A - class M M M NL - column 53, strong axis Y



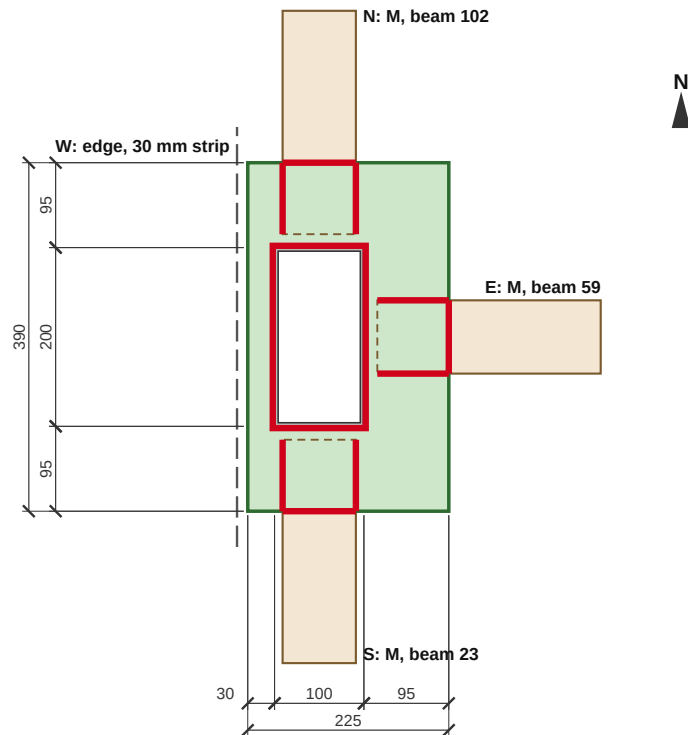
Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab.
(printed at about 1:7.5)

Joint 14 - type A - class M S M M - column 54, strong axis Y



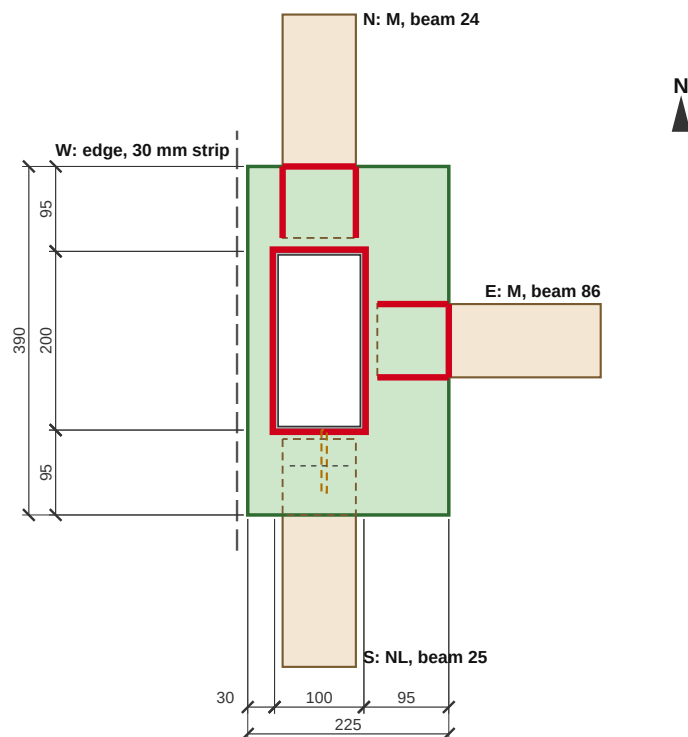
Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab.
(printed at about 1:7.5)

Joint 69 - type B - class E M M M - column 103, strong axis Y



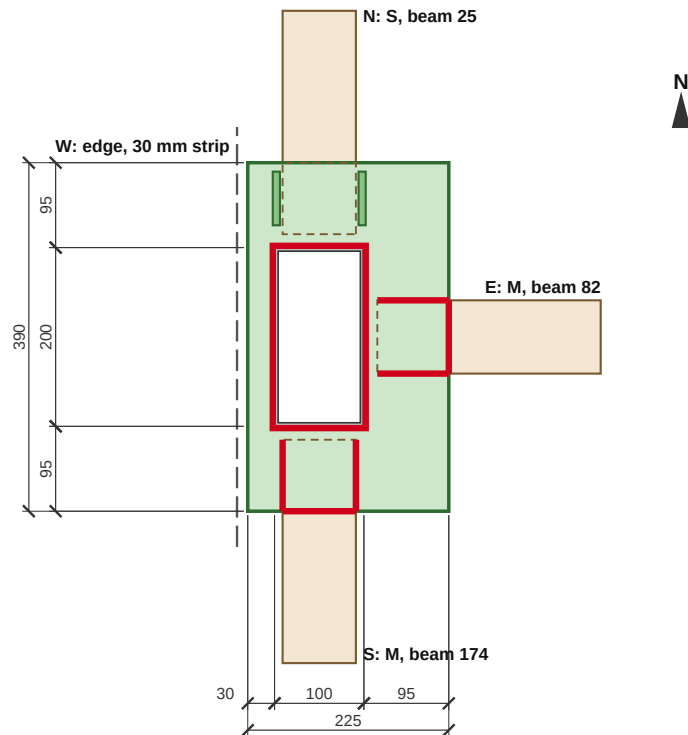
Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab. (printed at about 1:7.5)

Joint 71 - type B - class E M M NL - column 104, strong axis Y



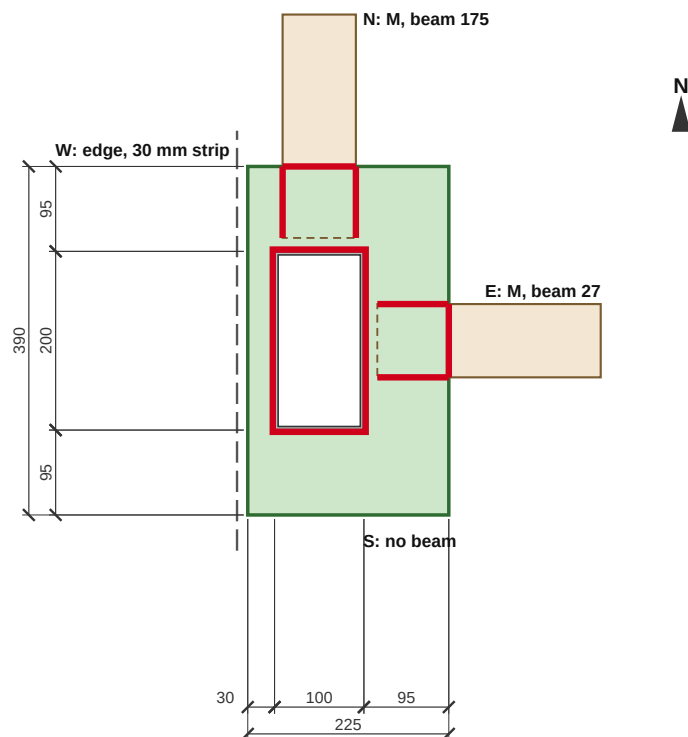
Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab. (printed at about 1:7.5)

Joint 72 - type B - class E S M M - column 105, strong axis Y



Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab. (printed at about 1:7.5)

Joint 73 - type B - class E M M N - column 106, strong axis Y



Cap shown; beam parts under the cap dashed. Red: welds (flange 3-sided, collar around the column). Green: stability plates. Orange: shear tab. (printed at about 1:7.5)